Chapter 2
Land Characteristics of the Amazon Region and Its Current Challenges

Verônica Garcia Donoso

Abstract The Amazon biome is a crucial element of the world's ecosystem. Its land characteristics encompass a complex system of rivers, streams, and wetlands along with a diverse array of vegetation and forest types. This complex ecosystem, which is characterized by rich biodiversity, is currently facing challenges related to land use changes. The objective of this chapter is to enhance awareness of the Amazon biome by presenting information about the land characteristics of the Amazon Basin and an analysis of the most significant challenges of the region, specifically addressing the state of Rondônia, in the Brazilian Amazon. It was developed using a multi-method approach, combining bibliographic research and mapping. The results indicate that the most significant drivers of deforestation are the expansion of cattle ranching and agricultural development. Moreover, the construction of infrastructure and the arrival of settlers through colonization projects have been linked to deforestation in the state of Rondônia. The conclusions present reflections on the potential strategies for a more sustainable future.

Keywords Amazon Rainforest · Land Characteristics · Deforestation · Agribusiness

2.1 Introduction

The Amazon is the world's largest tropical forest, boasting exceptional biodiversity. It is a crucial ecosystem globally, influencing the global carbon cycle and therefore climate change as well as the global hydrological systems that support the climate and rainfall for the South American continent. The Amazon region faces the challenges of an unplanned expansion of agriculture and cattle ranching and of illegal activities related to the exploitation and extraction of natural resources, such as

V. G. Donoso (✉)
Federal University of Santa Maria (UFSM), Cachoeira do Sul, Rio Grande do Sul, Brazil
e-mail: veronica.donoso@ufsm.br

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mining, timber and its products. As human activity pressure on the biome increases, a tipping point is becoming increasingly likely. This means that a situation could trigger a large-scale system change (Bottino et al., 2024; Wunderling et al., 2024; Lovejoy & Nobre, 2018).

The global magnitude of forest degradation and deforestation has a high cost for biodiversity, and its restoration is complex. Much research has been carried out on biodiversity and ecological restoration to mitigate the effects of deforestation in the Amazon region, much of it related to protected forest areas outside urban areas (Cavalcante et al., 2022; Leite-Filho et al., 2021; de Solar et al., 2016; Falcão et al., 2015; Nárgila et al., 2013). With regard to urban areas, much research was published recognizing that urban green spaces serve as habitat for wildlife and provide support for ecosystem services (Macedo et al., 2018a; LaPoint et al., 2015; Williams et al., 2015; Vergnes et al., 2013; Sushinsky et al., 2013).

Despite the acknowledged importance of forests and urban green areas, numerous questions remain regarding the optimal relationship between nature and human interference as well as the ecological planning strategies necessary to ensure high-quality living conditions for all species. This is particularly challenging in regions with complex natural ecosystems, such as the Amazon.

The objective of this chapter is to increase awareness of the Amazon biome, which has been through an extended period of time of deforestation and land degradation due to human activities such as logging, fires and agricultural practices.

It is divided into four sections. The first provides an introduction to the characteristics of the Amazon biome. The objective of this first section is to provide readers from diverse geographical and cultural backgrounds an overview of the Amazon Basin (AB) natural characteristics. Readers who are more familiar with the Amazon region may also find this overview useful. The second section will present an analysis of the most significant challenges currently facing the Amazon region. The third section will present the specific challenges facing Rondônia state, in the Brazilian Amazon. The concluding section of the chapter discusses sustainable paths for the future.

2.2 Methodology

In this chapter, an integrated multi-method approach is employed, combining particularly bibliographic research and mapping.

The Brazilian Amazon Basin maps were created using data from rivers and streams gathered by the Brazilian National Water and Basic Sanitation Agency (ANA, 2022). The topographical map analysis drew on data from the “Brazil em Relevo” initiative, which is a project of the Brazilian Agricultural Research Corporation (Embrapa). This project employs images from the Shuttle Radar Topography Mission (SRTM), collected by the National Imagery and Mapping Agency (NIMA) and the National Aeronautics and Space Administration (NASA) in 2000, to generate digital elevation models through the use of the SAR (interferometry) method. The data is accessible in raster format on the Brazilian project website (de Miranda, 2005).

To map the land use and land cover (LULC) from Rondônia state, database from the MapBiomas (2021), Collection 6.0, were used, combined with data from the Brazilian Institute of Geography and Statistics (2019, 2022).

2.3 Land Characteristics of the Amazon Region

The Amazon Rainforest covers an area of 944 million hectares, representing 47% of South America. It is shared by nine countries, with Brazil being the largest holder, accounting for 61.9% of the total area (521.9 million hectares). The Amazon Rainforest is Brazil's largest biome, covering 49.3% of the country's territory (Fig. 2.1).

The Brazilian Amazon biome is present in nine states: Acre, Amapá, Amazonas, Maranhão, Mato Grosso, Pará, Rondônia, Roraima, and Tocantins. Amazonas has the largest share of the biome, while other states also contain other biomes in their territories, such as the Cerrado and the Pantanal in Mato Grosso. Rondônia is notable for its extensive coverage of the Amazon biome, with 99% of its territory encompassed by this biome. In comparison, Mato Grosso has 54% of its territory covered by the Amazon biome, while Maranhão has 34% and Tocantins has 9%, as can be seen at Fig. 2.2 (IBGE, 2004).

In terms of natural characteristics, the Amazon Basin (AB) encompasses a wide variety of different soil types, reflecting the rich geologic diversity of origins and geomorphological processes (Quesada et al., 2011). The basin's relief is predominantly flat, with higher altitudes observed in the northern Amazonian mountains and the eastern slope of the Andes in the international Amazon region. In the Brazilian part of the Amazon region, the highest altitudes can be observed at Roraima state (Figure 2.3). The basin is characterized by a complex system of rivers, streams, and wetlands, which are irrigated by a high annual precipitation. This precipitation is distributed unevenly throughout the year, resulting in two distinct weather seasons: a rainy season and a dry season.[1] This variation in precipitation patterns has a significant impact on the hydrology, water, and soil fertility of the basin.

The AB is home to four of the world's largest river systems in terms of water discharge and drainage. Wittmann and Junk (2016) provide a comprehensive

[1] In the Brazilian Amazon, the dry season, which is defined as the period between June and November, is characterized by a monthly rainfall range of 42–162 mm. The period of rainfall is from December to May, with a monthly rainfall of 211–300 mm. The rainy season accounts for approximately 75% of the annual rainfall. A pioneering study of the two Amazonian weather seasons was conducted in 1984 (Ribeiro & Adis, 1984). In 2009, a study on the spatiotemporal variation of rainfall in the international Amazon was conducted by researchers from various institutions and published in a climatology journal. The study gathered data on precipitation from 1964 to 2003. Despite the limitations imposed by the uneven distribution of weather stations, the study contributes to the existing knowledge on rainfall in the Amazon basin, providing additional insights. These include the observation of a decline in rainfall associated with changes in the ocean and atmosphere, as well as a potential correlation with deforestation (Espinoza Villar et al., 2009).

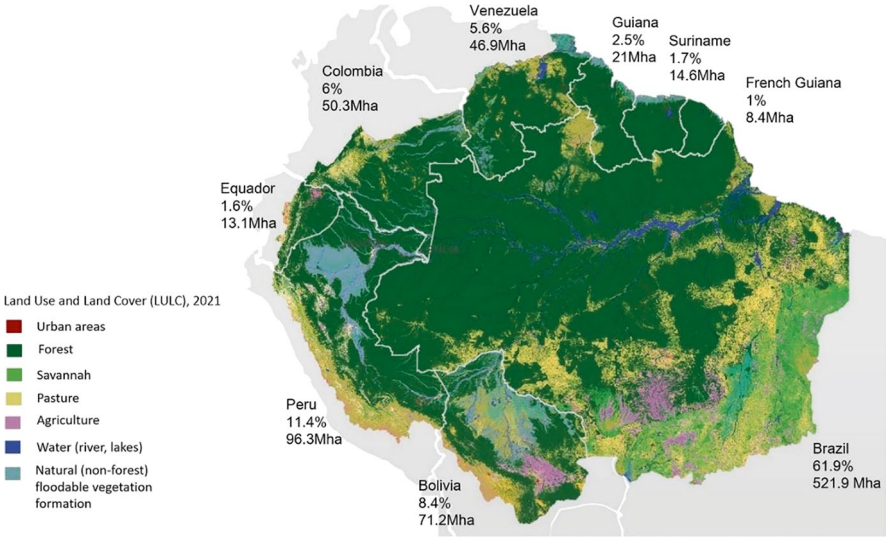

Fig. 2.1 International Amazon Biome—percentage in each country. (Source: Elaborated by Donoso, using data from Mapbiomas, 2021)

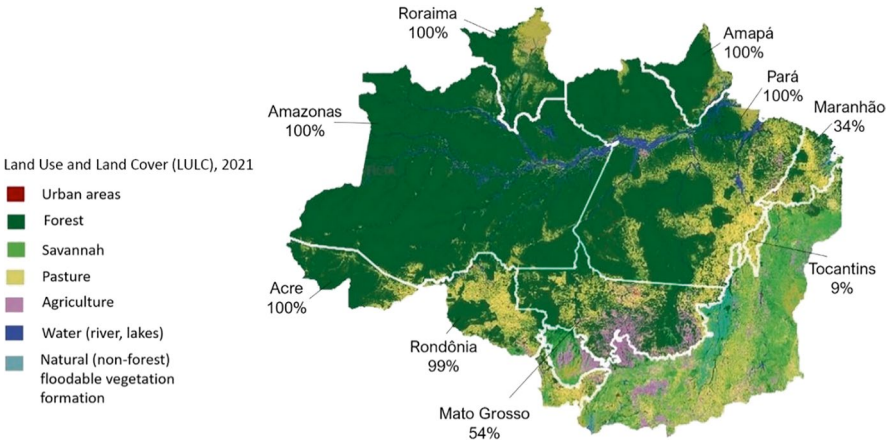

Fig. 2.2 Brazilian Amazon Biome—percentage in each state. (Source: Elaborated by Donoso, using data from Mapbiomas, 2021; IBGE, 2004)

account of the Amazon wetlands, incorporating an analysis of biodiversity. The authors describe the various types of wetlands that occur in accordance with the natural conditions, including seasonal floodplains along river systems, interfluvial wetlands of episodically flooded riparian areas along streams, permanent flooded swamps, and hydromorphic savannas. Coastal wetlands encompass mangroves and tidal floodplains. Wolfgang Junk, together with Maria Piedade, also published a book chapter about the ecology of the South American Wetland Forests, detailing the distribution of species in the Amazon (Junk & Piedade, 2010).

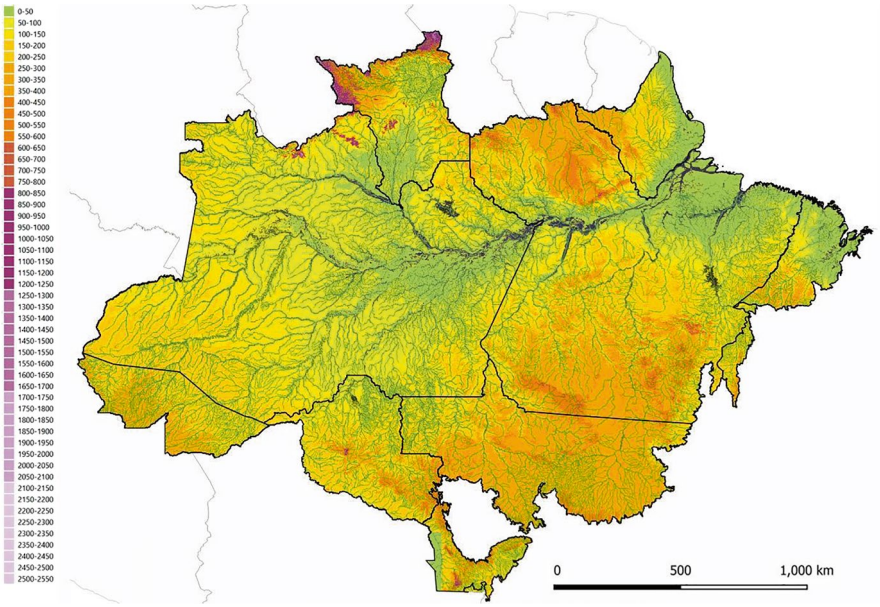

Fig. 2.3 Brazilian Amazon Basin: River system and Topography. (Source: Elaborated by Donoso, using data from ANA (2022) and de Miranda (2005))

This diversity of landscape scenarios is also present in the vegetation, and several forest types can be found in the apparently homogenous rainforest. The variations of forest types are related with the water cycle and soil, with the presence of *terra firme* forest, in non-floodable areas, and the floodable habitats or alluvial forests, known in Brazil also as *igapós, igarapés* and *várzeas* (Gonçalves et al., 2022).

Terra firme is the forest type with the highest tree alpha-diversity. In contrast, forests on *igapó*, white sand, *várzeas*, and other floodable habitats have a lower tree alpha-diversity, which is related to the poor nutrient status of the soil and the flooding periods (ter Steege et al., 2023). Additionally, other research has demonstrated the correlation between these inundation patterns and the assemblages of fauna and flora (Laranjeiras, Naka and Cohn-Haft, 2019; Barnett & Jucá, 2018; Montero et al., 2014; Parolin et al., 2013).

The biodiversity and connectivity of freshwater and terrestrial ecosystems are experiencing a significant decline due to the expansion of anthropogenic activities, including deforestation, pollution, and the construction of dams and waterways. Although the Amazon Basin is the subject of considerable scientific attention, there is still a need to extend the protection of this important ecosystem. Furthermore, it is important to recognize the importance of the freshwater ecosystem, which is a vital component of the basin (Donoso & Reicher, 2024; Castello et al., 2013).

The variation of fauna in the Amazon region is still not easy to measure by science. It is home for many endemic species, some endangered, but also many still unknown. Some identified species indicate potential to generate global human benefits, such as the case of the Fer de Lance venom, a tropical viper that help people around the world control hypertension (Lovejoy, 2019).

The Amazon Basin is a region of great cultural and social diversity, the result of a complex historical development. Upon the arrival of Europeans in the Amazon in the sixteenth century, they encountered a forest populated by numerous indigenous tribes exhibiting a rich cultural diversity. The Brazilian archaeologist Eduardo Neves (2022) asserts that the earliest known records of human habitation in the Amazon region date back approximately 14,000 years. He also estimates that approximately 5 million indigenous people inhabited the Amazon region prior to the European invasion.

The social diversity of the AB can be better understood when divided into two main categories: the forest people and traditional communities, and the urban population. This classification allows for a more nuanced understanding of the social groups that inhabit the forest. Among the first category are indigenous peoples, quilombolas, riparian communities and fisherfolk. In addition to the forest people, numerous cities and urban nuclei can be found in the Amazon Basin. This diverse social reality is the result of various historical moments, including missionary work, rubber extraction activities, mining, the promotion of settlement occupations mostly by the federal government, the creation and expansion of road infrastructures and communication infrastructure, the exploitation of water systems for energy production, the emergence of illegal activities such as logging, fires and mining, and so forth. This resulted in the social and economic development of the region, which in turn led to profound alterations in the ecosystems.

Figure 2.4 shows the topography of the Brazilian AB region, including urban areas. The map illustrates a more pronounced degree of urbanization in the coastal

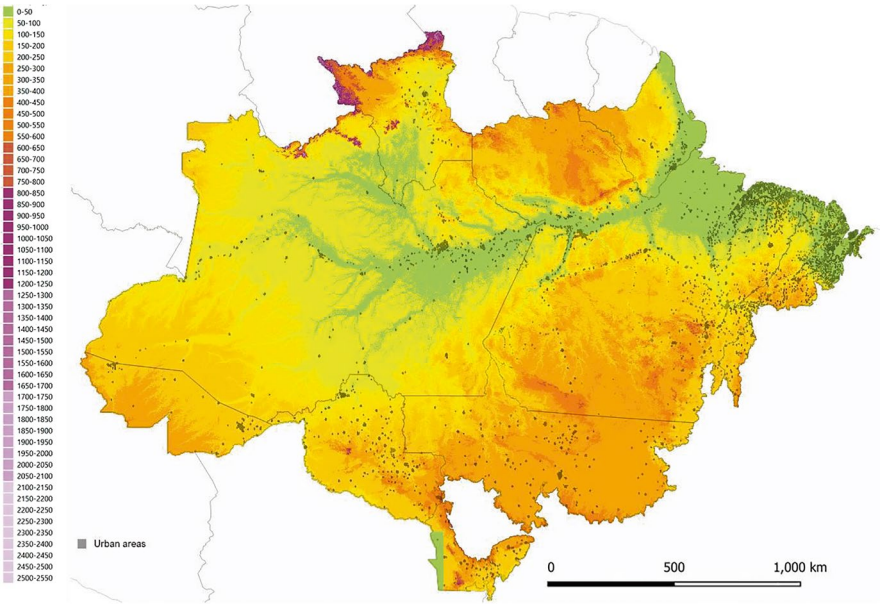

Fig. 2.4 Brazilian Amazon Basin: Topography and urban areas. (Source: Elaborated by Donoso, using data from ANA (2022), de Miranda (2005) and IBGE (2020))

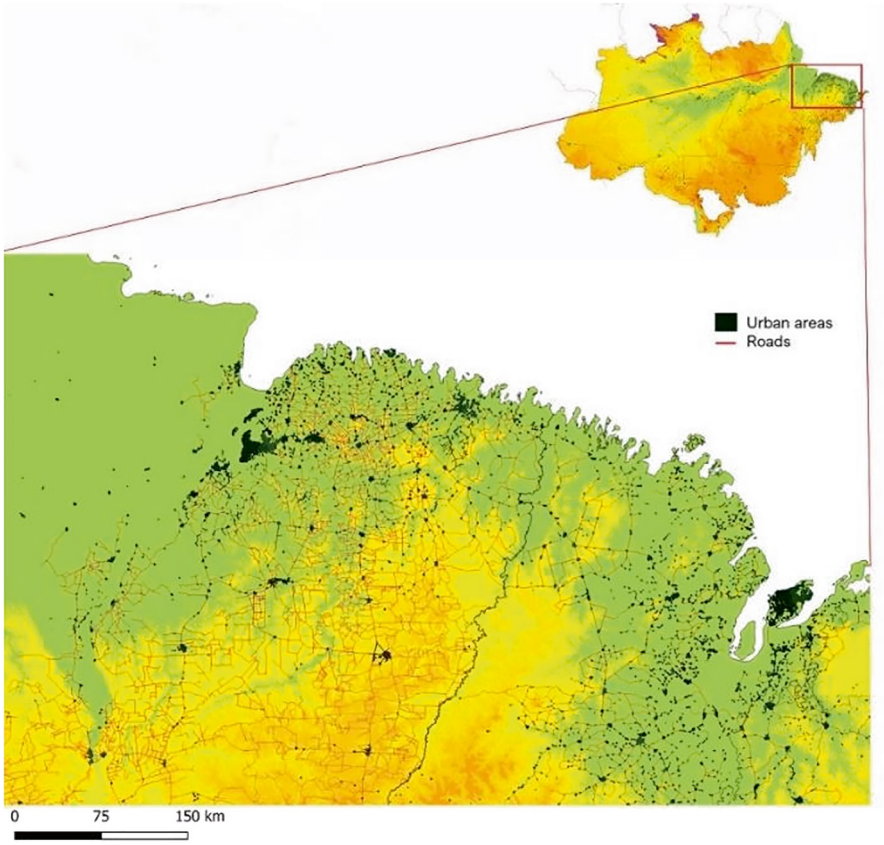

Fig. 2.5 Urban expansion of Pará and Maranhão coastal area. (Source: Elaborated by Donoso, using data from ANA (2022), de Miranda (2005) and IBGE (2020))

regions of Maranhão and Pará. This urban expansion is not a single urban agglomeration, but rather a collection of small urban nuclei dispersed across sprawled territories. This is illustrated in Figure 2.5.

2.4 The Amazon Region Current Challenges

Over the last 40 years, the Amazon has been heavily deforested and continues to deforest rapidly. From 1985 to 2020, the Amazon lost 74.6 Mha of native vegetation, with Brazil accounting for almost 60% of the deforestation (Mapbiomas, 2021).

Deforestation is one of the greatest challenges facing the Amazon rainforest, and Researchers have pointed out that deforestation is putting the Amazon in danger of reaching an irreversible tipping point. The ongoing expansion of deforestation is also affecting the local climate, intensifying drought regimes across the region (Donoso & Reicher, 2024; Anderson et al., 2018).

As a pivotal element of the Earth's climate system, the Amazon climate is intricately linked on a global scale through ocean and atmospheric circulation, as well as other interconnected systems. The interconnections between subsystems are referred to as teleconnections. For many years, researchers have been focusing on these teleconnection events, noting that the future global climate will depend on a more effective interaction between human activity and the natural environment (Wunderling et al., 2024; Kravtosov et al., 2018).

The teleconnection hypothesis offers a means of understanding how climate-related events affect other regions. For example, a significant change in sea surface elevation will result in interactions and atmospheric processes in other regions around the globe. This implies that it is not only important to protect the Amazon biome for the world's climate, but the planet must also be cared for in order to protect the Amazon. In other words, deforestation in the Amazon affects global events, and global events affect the Amazon region. This implies that the current challenges of the Amazon regions are not limited to the Amazon itself, but also related with our actions and human impacts around the globe. As an example, the potential for more frequent El Niño Southern Oscillation (ENSO) event could influence precipitation patterns in the Amazon region, leading to the emergence of extreme weather conditions such as droughts in certain regions of South America (Duque-Villegas et al., 2019).

Considering the local impacts of land use change in the Amazon region, the main drivers of deforestation are the expansion of agriculture and cattle ranching, illegal logging, fires and illegal and unregulated mining. This is a well-known problem, but it is also a political issue, with a lack of law enforcement to curb illegal activities, a lack of governance and poor decision making in the management of natural resources.

The 'Amazonia Under Pressure' atlas highlights some of the greatest challenges facing the Amazon. Published by RAISG in 2020, it points out the important role of cattle ranching in the destruction of the forest. It also focuses on the crucial role of infrastructure in driving deforestation, as the loss of vegetation is directly linked to the various access roads that cut into the forest.

Deforestation in the international Amazon is predominantly caused by anthropogenic activities, specifically pasture and agriculture, which account for 99% of the deforestation. Other anthropogenic causes, such as urbanization, only account for 1% of deforestation. Despite this, urban areas have expanded by 0.6 million hectares during the period from 1985 to 2021, representing a 152% increase. In contrast, mining activities have caused an expansion greater than 0.5%, with a 1000% increase over the same period (MapBiomas Amazônia, 2022).

About 12% of the Brazilian forest area (44 Mha) disappeared from 1985 to 2021, and most of the deforestation can be linked to cattle ranching, followed by small

percentages of deforestation due to illegal mining and urban expansion. Among the Brazilian Amazon states, Pará and Rondônia stand out in the conversion of forest to agriculture or pasture (Donoso et al., 2024; Aguiar et al., 2012; Mapbiomas, 2022; Recanati et al., 2015). This book's focus on Rondônia is motivated by the necessity to discuss sustainable urban growth strategies for the region.

2.5 Rondônia Current Challenges

The state of Rondônia is positioned in the southwestern part of the Brazilian Amazon and was originally almost entirely rainforest. In 2021, compared to 1985, the natural forest vegetation area in the state reduced 66%. As shown in Figure 2.6, the natural forest vegetation was 87.10% of the state's area in 1985. In 2021, the natural forest vegetation was 57.58%. On the other hand, the pasture area, which in 1985 occupied 8.21%, increased to 36.34% in 2021 (Mapbiomas, 2021).

Rondônia state origin was the result of an international agreement between Brazil and Bolivia (Tratado de Petrópolis, Nov. 17, 1903) that allowed the construction of the Madeira Mamoré Railroad, at the beginning of the 20th century. That railroad aimed to connect Bolivia to the Atlantic Ocean to transport rubber production. This infrastructure enabled the creation of some cities, such as the capital Porto Velho in 1908 (Morelato & Cardoso, 2020).

Over the past several decades, the Brazilian State of Rondônia has been the destination of many rural migrants drawn from Brazil's middle southern regions by massive government colonization projects. Factors such as explosive population

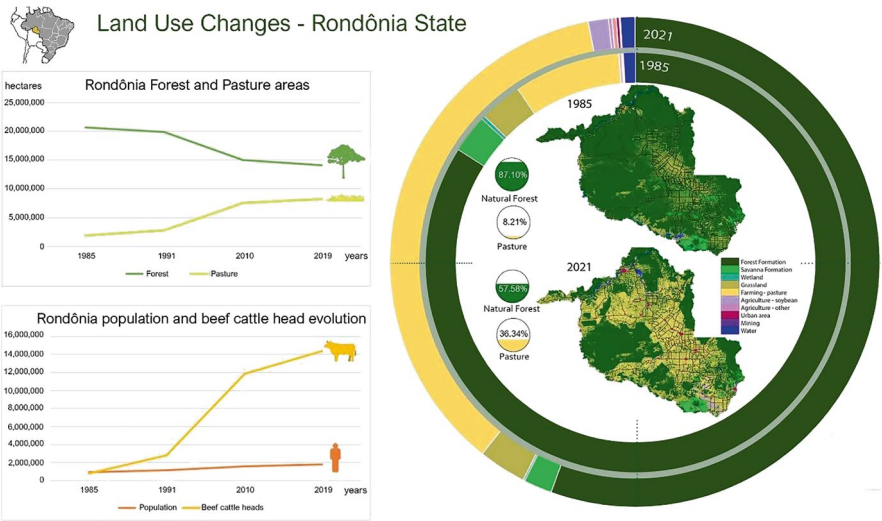

Fig. 2.6 Land use changes in Rondônia state. (Source: elaborated by Donoso. Data sources: Mapbiomas (2021), IBGE (2019, 2022))

growth, logging, mining, small scale farming, and ranching have synergistically fuelled deforestation in the state (Franco, 2019; da Cunha & Moser, 2010; de Oliveira, 2010). This state is an example of an almost irreversible tipping point of deforestation of the Amazon rainforest (Figure 2.6).

Rondônia is still undergoing a period of major agricultural expansion, currently with the growth of soy plantations.[2] Considering the urban areas, in 1985, these occupied only 0.08% of the area of Rondônia state, increasing to 0.21% in 2021.

2.6 Conclusions

It is of great importance to develop long-term, sustainable ways in which humans live and relate to nature. It thus follows that investment must be made into the development of sustainable bio-economies and the production and consumption of sustainable foods, in order to feed the increasing global population. Furthermore, it is essential to implement sustainable solutions to improve the way in which we live and plan our cities. Moreover, it is vital to enhance protection and awareness of the world's most significant biomes.

A number of environmentally friendly measures are needed. These include the establishment and management of protected areas, the creation of sustainable development plans, efficient public policies, and better planning of urban and infrastructure expansion as well as the strengthening of control and monitoring of forest areas. In fact, researchers have shown that a 1% increase in fines imposed by Brazil's Institute of the Environment and Renewable Natural Resources (IBAMA) reduced deforestation by 0.5% (Amaral, 2023b).

The importance of action at the local level cannot be underestimated. In the case of Brazil, there is a problem in controlling deforestation because permits for the suppression of native vegetation can be issued by state environmental agencies, which makes it difficult to take national and global action to control deforestation. One solution would be for states to focus on using already deforested areas to the detriment of further expansion (Gabriel, 2023).

It is important for the population to be made aware that the economic benefits of the forest outweigh those of exploiting it. In addition to agricultural productivity, regenerative agriculture and urban productivity in the Amazon can slow deforestation and stimulate regional and national development. This information is contained in the economic memorandum "A Balancing Act for Brazil's Amazonian States," published by the World Bank (Hanusch, 2023). The document also highlights that the standing forest is worth seven times more than it is exploited: the "standing" forest is estimated to be worth more than US$317 billion per year. This is the general regional and global public value for environmental protection. These values

[2] The agricultural production increased on the state with the soybean plantations from the 1990s onwards, reaching 1.34% of the state's area in 2021.

have been calculated on the basis of the ecosystem services provided by the forest, such as the generation of rainfall necessary for agriculture, soil protection against erosion and fires, carbon storage, and biodiversity conservation. On the other hand, the value of using forests for timber, mining, and agriculture generates between US$43 billion and US$98 billion annually (Amaral, 2023a).

Brazil faces the impact of colonialism and imposed policies on its development trajectories, which have resulted in marked ethno-socio-economic imbalances and marginalization (Donoso & Queiroga, 2023). In places with such mixed challenges, urban and landscape design is neither climate-resilient nor appropriate to the surrounding biodiversity (Caputo et al., 2019).

In an approach to sustainable urban development, green spaces are a key element. In combination with biodiversity, it is a step toward an ecological approach. This means recognizing that urban green spaces can contribute to the conservation of biodiversity and the principles of landscape ecology. Among other things, landscape ecology theory defines the pathway-corridor-matrix spatial pattern as a landscape component (Forman & Gordon, 1986). Together with a systemic view of the importance of open spaces (Macedo et al., 2018a, 2018b), it can support the conservation and restoration of biodiversity in urban areas.

Urban green spaces provide habitats for a wide range of wildlife. However, the size of urban areas, but also urban morphology and the availability of open spaces and green areas in cities can affect biodiversity. There are still many questions to be answered about how cities can be planned to provide good conditions for people while protecting biodiversity.

In fact, there is evidence that an increase in urban green spaces, including backyards, and functional links between them is important to avoid or mitigate the negative effects of urbanization on wildlife (Lopucki et al., 2020; Sushinsky et al., 2013; Vergnes et al., 2013; Williams et al., 2015; LaPoint et al., 2015; Matthies et al., 2017).

The importance of nature-based solutions (NbS) in contributing to sustainable development and mitigating climate change is also highlighted (Kinol et al., 2023; Sisay et al., 2023; Roggema & Tillie, 2022). Some researchers have already found that to ensure the biodiversity of native wildlife in cities, action should be taken in the early stages of urban development. They also highlight the fact that the theories of sustainable design for small cities are still inadequate (Lopucki et al., 2020).

This may not sound new or innovative to some critics, but in areas like Rondônia State, basic actions and knowledge still need to be implemented to protect the future of the Amazon region. In fact, this pathway approach could be interesting for many Global South contexts. It is important to emphasize that applications of NbS carried out in the context of the Global North provide many valuable concepts and methodologies, but they should not be readily transferred to contexts characterized by unplanned and unregulated decisions and actions, and also to contexts of extreme inequality, such as the Amazon region.Many cities in the state of Rondônia have been planned with high-density blocks and few or no urban trees. In addition, many cities have urbanized streams, known as igarapés, or have polluted them. Urban design, planning, and legal measures to increase green spaces and their connectivity

in the urban fabric could support biodiversity in Amazonian cities, if combined with social and political support and environmental education.

Nevertheless, ecological planning still faces difficulties in becoming a real strategic priority in territorial planning, despite the urgency of the topic. In light of these conceptual gateways, promoting cities where there is balance between natural and human processes is every day more urgent.

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